

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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Football Match Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Football Match Manager is a data-centric system designed to manage football match lifecycles, team registries, and league structures. It implements a relational pattern to track match states (upcoming to finished), score validations, and user-match following relationships. The system focuses on strict data integrity and optimized query performance through specific indexing strategies for match filtering and user authentication.

1.2 Maturity Assessment

The project is currently in **REFINEMENT**. While the data model is structurally complete with a high completeness score, it lacks critical business context. The absence of defined Goals & Objectives and Scope boundaries creates a high-severity gap that prevents full execution readiness, as the functional purpose of the application remains implicit rather than explicit.

1.3 Technical Stack

- **Primary Identifiers:** UUID (128-bit)
- **Temporal Standard:** UTC
- **Data Types:** Enum (Match Status), DateTime, URL, Boolean, Integer, String

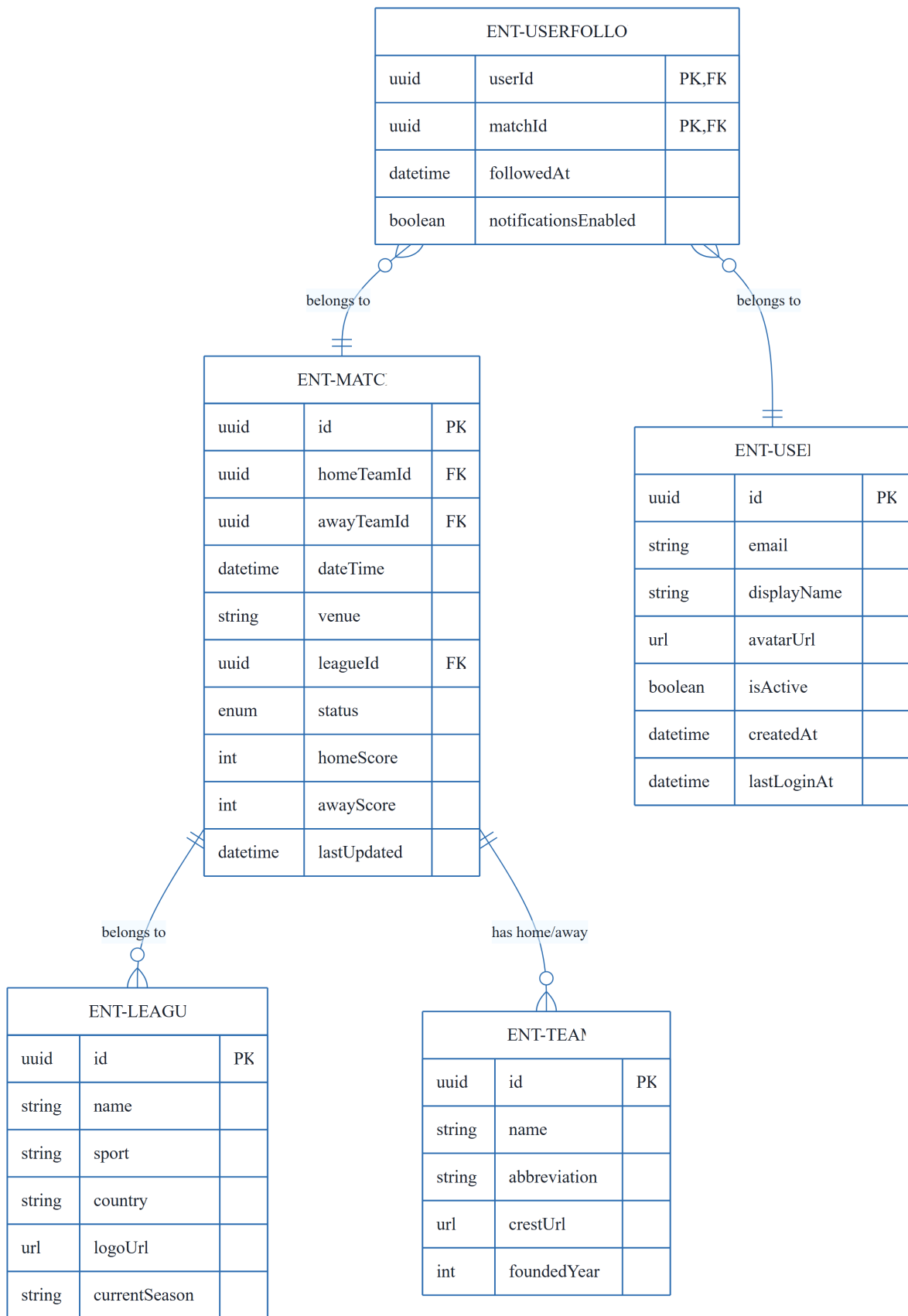
1.4 Architectural Constraints

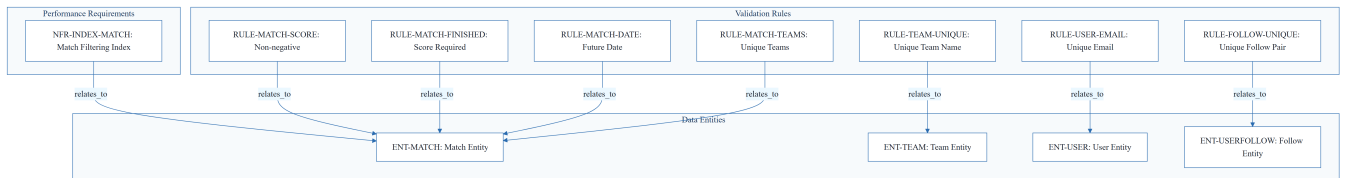
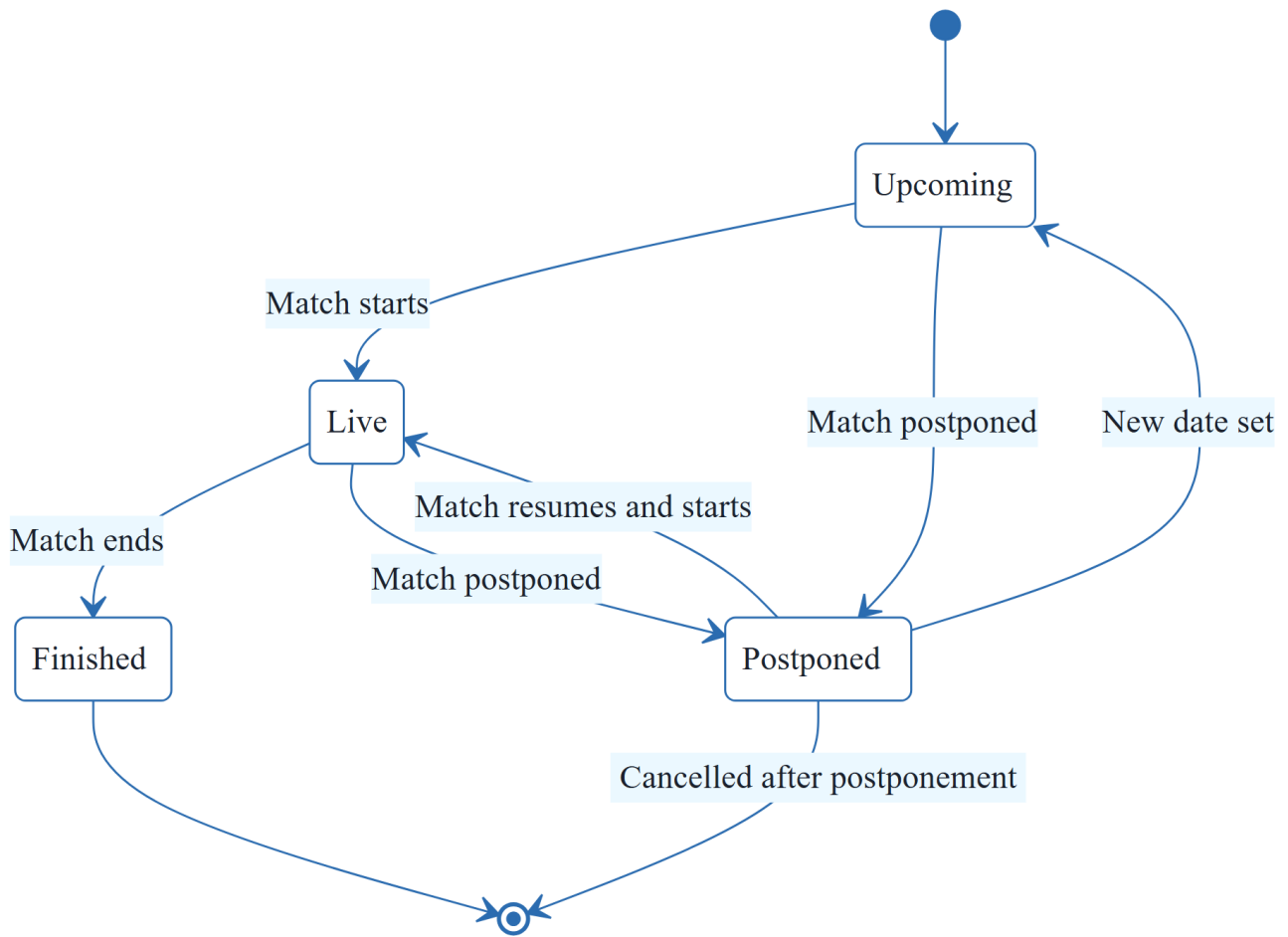
- **Match Integrity:** homeTeamId must not equal awayTeamId.
- **Temporal Logic:** Match dateTime must be in the future if status is 'upcoming'.
- **State-Based Requirements:** homeScore and awayScore must be non-null when status is 'finished'.
- **Value Constraints:** homeScore and awayScore minimum value: 0.
- **Uniqueness:** Team name must be globally unique; User email must be unique.
- **Data Limits:** Team abbreviation max 5 characters (unique per league); Team name max 100 characters; User displayName max 50 characters; Match venue max 200 characters.
- **Relational Constraints:** Composite unique constraint on UserFollow (userId, matchId).
- **Domain Range:** Team foundedYear must be between 1800 and the current year.

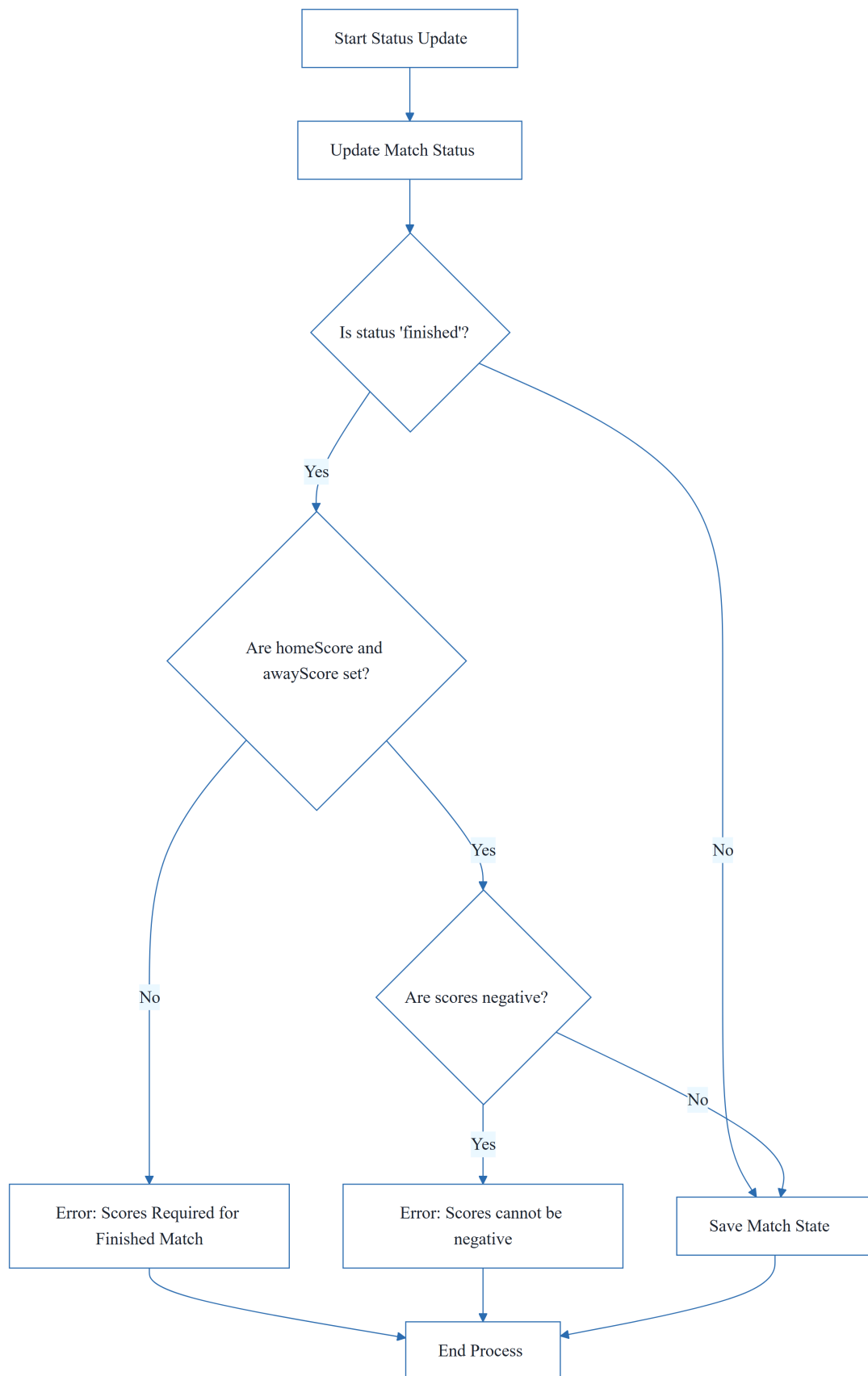
1.5 Critical Dependencies

- **Entity Dependencies:** Match entity has strict foreign key dependence on Team and League entities.
- **Relational Dependencies:** UserFollow entity has strict foreign key dependence on User and Match entities.
- **Performance Dependencies:**
 - Index on (leagueId, dateTime) for Match filtering.
 - Index on (status) for Match state filtering.
 - Index on (email) for User authentication.
 - Index on (userId) and (matchId) for UserFollow relational lookups.

2. Architecture Workflows

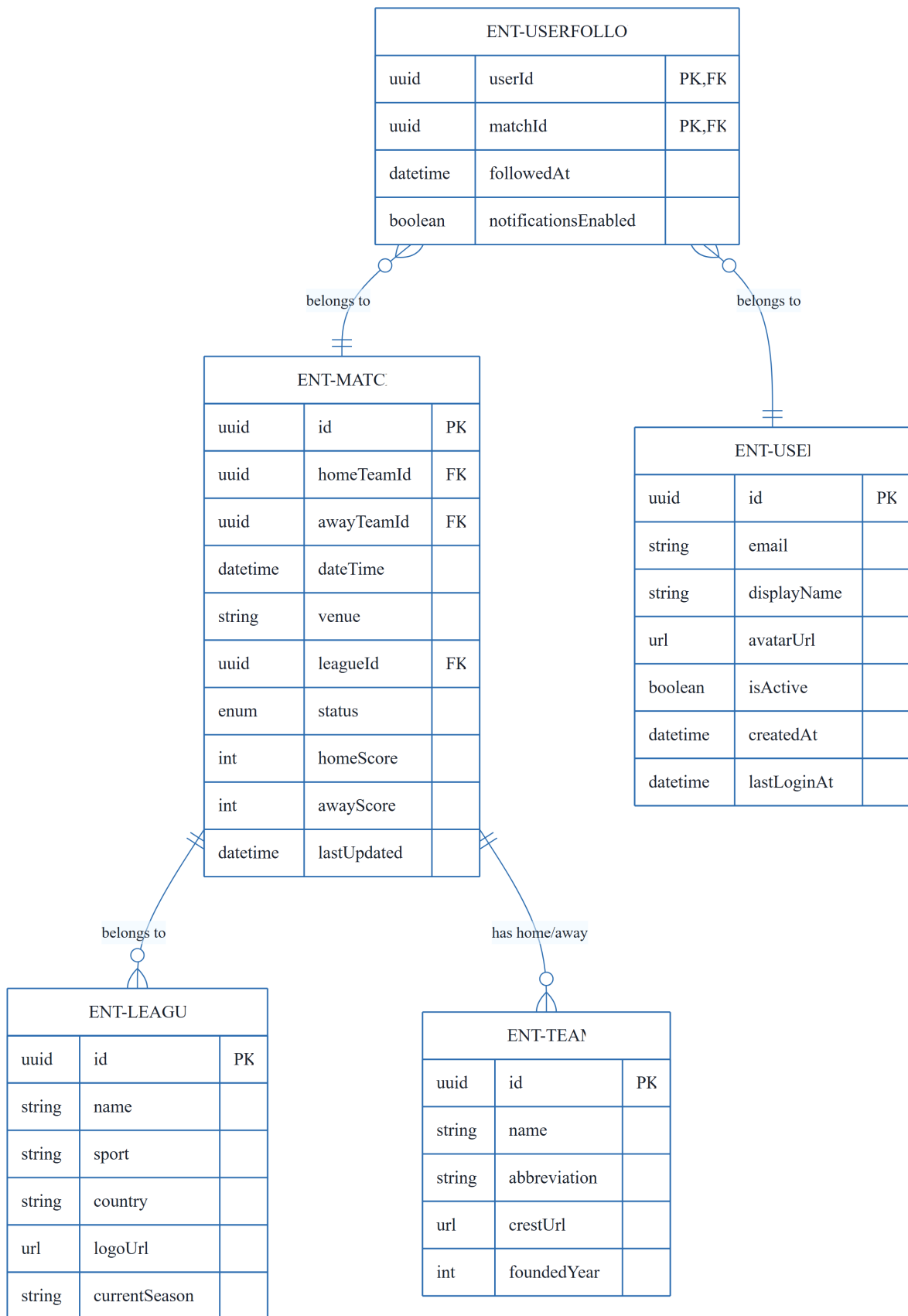




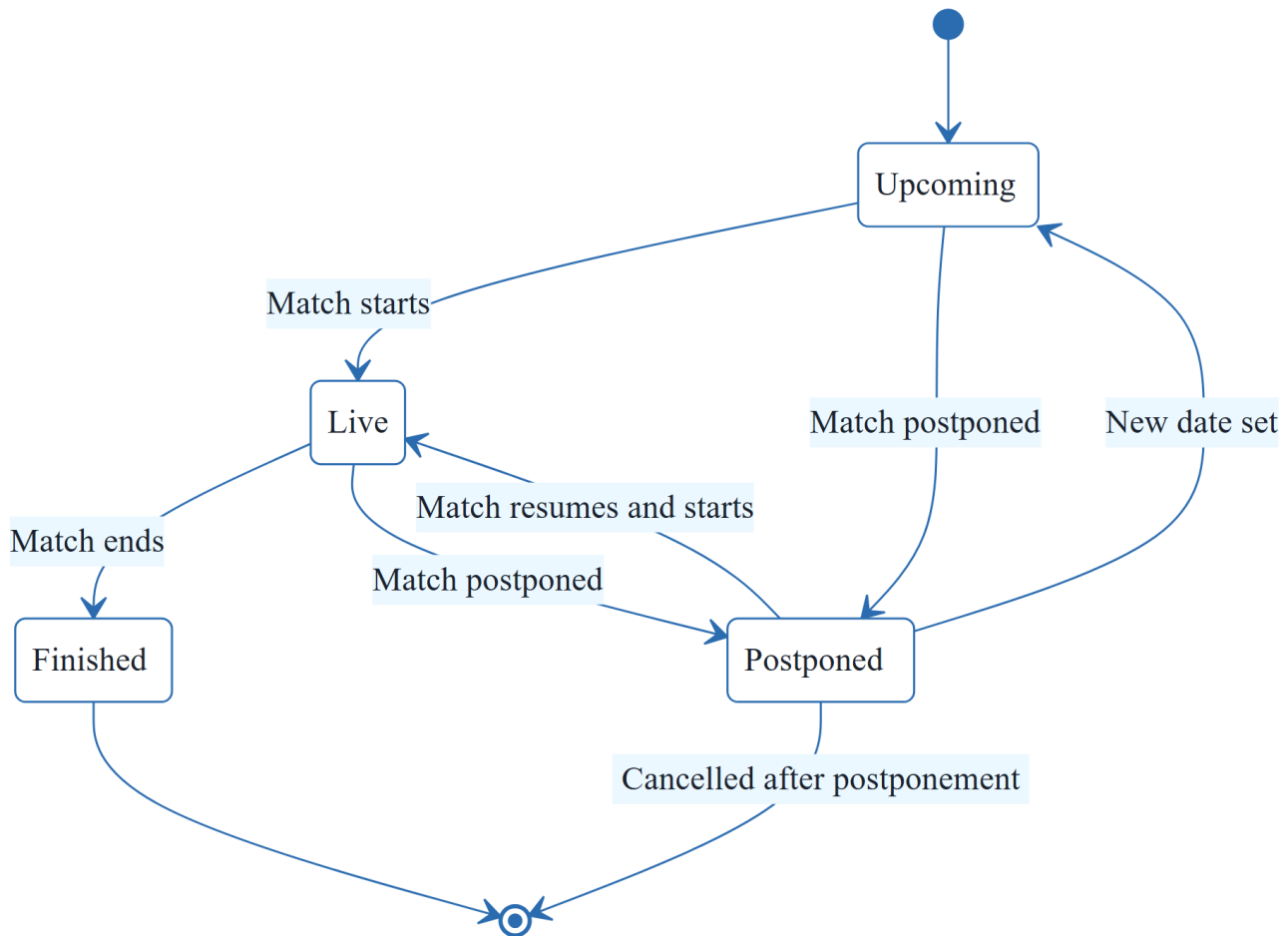


& Visual Diagrams

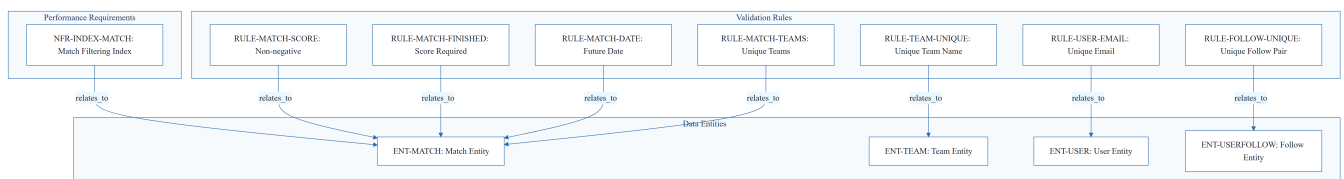
2.1 Data Model (ERD)



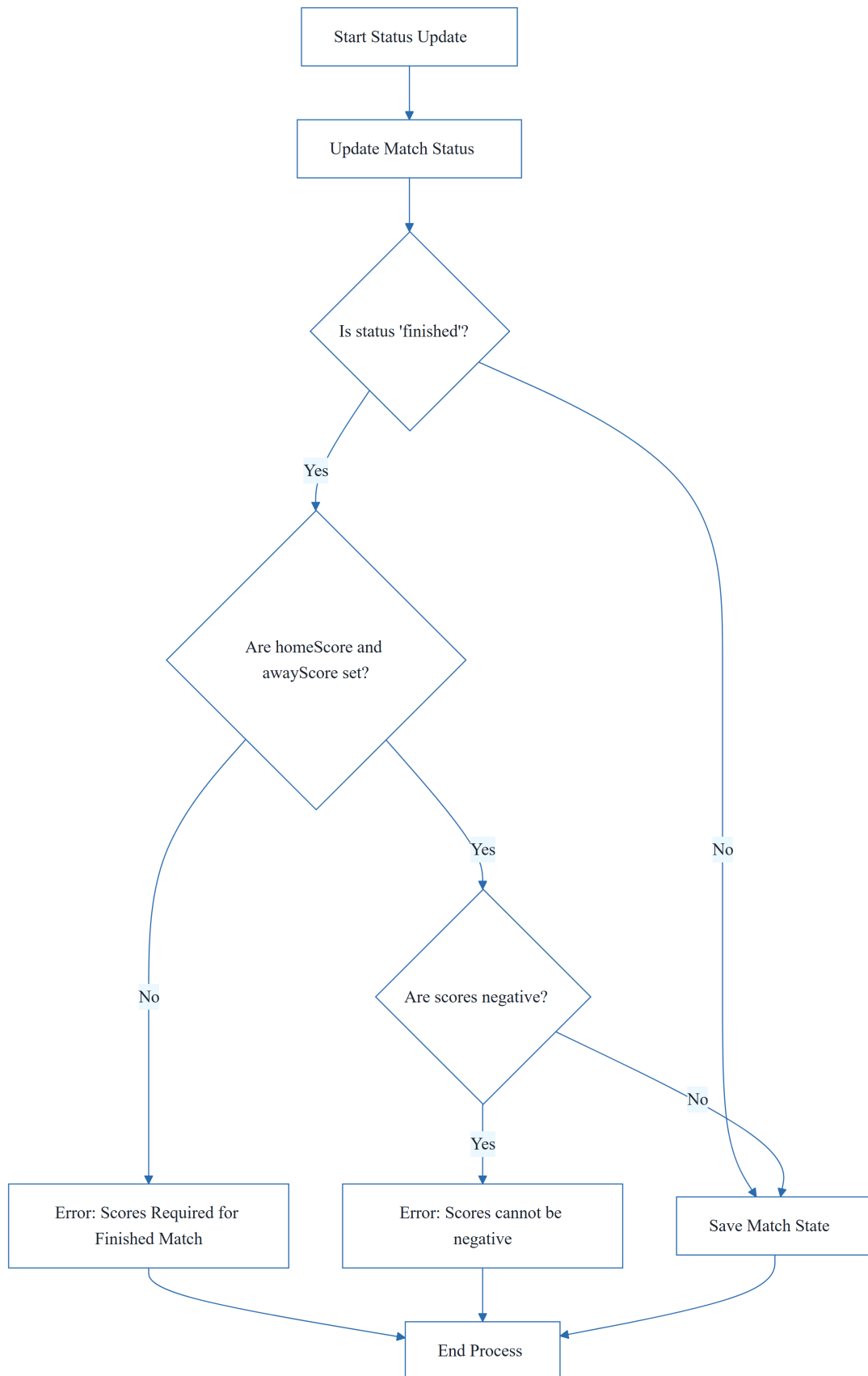
2.2 Match State Lifecycle



2.3 Data Validation & Constraint Traceability



2.4 Match Status Update Workflow



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Type	Description	Source Section
ENT-MATCH	Entity	Represents a football match with details on teams, date, venue, league, status and scores.	Match
ENT-TEAM	Entity	Represents a football team including name, abbreviation, crest and foundation year.	Team
ENT-LEAGUE	Entity	Represents a football league or tournament.	League
ENT-USER	Entity	Represents an application user with email and profile information.	User
ENT-USERFOLLOW	Entity	Represents the relationship between a User and a Match they are following.	UserFollow
RULE-MATCH-TEAMS	Constraint	A match cannot have the same team as both home and away.	Validation Rules
RULE-MATCH-DATE	Constraint	Match dateTime must be in the future for status 'upcoming'.	Validation Rules
RULE-MATCH-FINISHED	Constraint	When status changes to 'finished', both homeScore and awayScore must be set (non-null).	Validation Rules
RULE-MATCH-SCORE	Constraint	Scores cannot be negative.	Validation Rules
RULE-TEAM-UNIQUE	Constraint	Team name must be unique across all teams.	Validation Rules
RULE-USER-EMAIL	Constraint	Email must be unique.	Validation Rules
RULE-FOLLOW-UNIQUE	Constraint	A user can only follow a match once (Composite unique constraint on userId, matchId).	Validation Rules
NFR-INDEX-MATCH	NFR	Indexes on (leagueId, dateTime) and (status) for Match filtering performance.	Indexes (for performance)

3.2 Security Rules

- **Authentication:** User email must be unique and follow a valid email format (RULE-USER-EMAIL).
- **Credential Safety:** Passwords (where applicable) must meet complexity requirements (min 8 characters).
- **Account State:** User accounts are managed via an `isActive` boolean flag.

3.3 Data Models

ENT-MATCH

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
homeTeamId	UUID	Reference to Team	Required, exists
awayTeamId	UUID	Reference to Team	Required, exists, != homeTeamId
dateTime	DateTime	Match date and time (UTC)	Required, future if 'upcoming'
venue	String	Stadium or location name	Optional, max 200
leagueId	UUID	Reference to League	Required, exists
status	Enum	upcoming, live, finished, postponed, cancelled	Required, default: upcoming
homeScore	Integer	Goals scored by home team	Optional, min 0, default null
awayScore	Integer	Goals scored by away team	Optional, min 0, default null
lastUpdated	DateTime	Timestamp of last update	Auto-set on update

ENT-TEAM

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
name	String	Team name	Required, max 100, unique
abbreviation	String	Team abbreviation	Optional, max 5, unique per league
crestUrl	URL	URL to team crest/logo	Optional, valid URL
foundedYear	Integer	Year the club was founded	Optional, 1800 to current year

ENT-LEAGUE

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
name	String	League name	Required, max 100
sport	String	Sport type	Required, default: "football"
country	String	Country where league is based	Optional, max 100
logoUrl	URL	URL to league logo	Optional, valid URL
currentSeason	String	Current season identifier	Optional (e.g., "2023/24")

ENT-USER

Field	Type	Description	Validation
id	UUID	Unique identifier	Required, unique
email	String	User email address	Required, unique, valid email
displayName	String	Display name for the user	Required, max 50
avatarUrl	URL	URL to user avatar/image	Optional, valid URL
isActive	Boolean	Whether the user account is active	Required, default: true
createdAt	DateTime	Account creation timestamp	Auto-set on create
lastLoginAt	DateTime	Last login timestamp	Updated on login

ENT-USERFOLLOW

Field	Type	Description	Validation
userId	UUID	Reference to User	Required, exists
matchId	UUID	Reference to Match	Required, exists
followedAt	DateTime	Timestamp of following	Required, auto-set on create
notificationsEnabled	Boolean	Notification preference	Optional, default: false

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Goals & Objectives	HIGH	Define the business goals and the purpose of the Football Match Manager application.
Scope & Out-of-Scope	MEDIUM	Clarify what the system will and will not handle (e.g., player stats, ticket sales).
Open Questions & Uncertainties	LOW	List any unresolved technical or business decisions.

4.2 Remediation & Workflow

The project must transition from the **REFINEMENT** phase to **EXECUTION** by addressing the high-priority gaps. The primary focus should be the definition of the business purpose to ensure the data model aligns with the intended user experience and operational goals.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
Constraints	TECHNICAL_STACK	ENT-USERFOLLOW	The set of structural limitations, such as composite uniqueness on user and match identifiers, ensuring data integrity.
Cryptographic Hashing	TECHNICAL_STACK	RULE-USER-EMAIL	The required security mechanism for storing sensitive authentication credentials to meet complexity and safety standards.
DateTime	TECHNICAL_STACK	ENT-MATCH	The temporal data type used for scheduling events and recording audit timestamps.
MUN	BUSINESS_DOMAIN	ENT-TEAM	An example of a short-form identifier for a sports club, limited to five characters.
Match Validation	BUSINESS_DOMAIN	Validation Rules	The logic ensuring that opposing teams are distinct, dates are future-dated for upcoming events, and scores are non-negative.
Relationships	TECHNICAL_STACK	ENT-MATCH	The associative links between entities, such as the connection between a game and its participating clubs or league.

Term	Category	Context Anchor	Project Definition
Team Validation	BUSINESS_DOMAIN	RULE-TEAM-UNIQUE	The requirement that a club's name must be globally unique across the system.
UTC	TECHNICAL_STACK	ENT-MATCH	The standardized time zone used for all temporal records to ensure global synchronization.
UUID	TECHNICAL_STACK	ENT-MATCH	The 128-bit universally unique identifier used as the primary key for all major entities.
User Validation	BUSINESS_DOMAIN	RULE-USER-EMAIL	The set of rules enforcing unique email addresses and minimum character lengths for security credentials.
UserFollow	BUSINESS_DOMAIN	ENT-USERFOLLOW	The associative entity linking a person to a specific game they wish to monitor.
UserFollow Validation	BUSINESS_DOMAIN	RULE-FOLLOW-UNIQUE	The restriction preventing a single person from monitoring the same game multiple times.